

Asmita Pal

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Summary

Recent PhD graduate in Computer Architecture with experience in privacy-preserving traces, homomorphic encryption, and interconnect security. Skilled in performance modeling and privacy-preserving trace generation, with applications spanning general-purpose workloads, machine learning, and fully homomorphic encryption.

Education

PhD in Computer Engineering

University of Wisconsin - Madison

August 2018 – August 2025

MS in Computer Engineering

Utah State University

August 2016 – August 2018

B.Tech. in Electronics and Instrumentation Engineering

West Bengal University of Technology, India

July 2010 – June 2014

Publications

- J Kalamatianos, JB Kotra, A Pal: **Managing a Cache Using Per Memory Region Reuse Distance Estimation**, US Patent App 20240211407
 - Asmita Pal, Keerthana Desai, Rahul Chatterjee, Joshua San Miguel: **Camouflage: Utility-Aware Obfuscation for Accurate Simulation of Sensitive Program Traces**, published in ACM Trans. Archit. Code Optim. 21, 2, Article 36 (June 2024), 23 pages.
 - Mitali Soni, Asmita Pal, Joshua San Miguel: **As-Is Approximate Computing**, published in ACM Trans. Archit. Code Optim. 20, 1, Article 3 (March 2023), 26 pages.
 - Asmita Pal, Karthik Swaminathan, Subhankar Pal, Joshua San Miguel: **Characterizing Memory side channels in FHE applications** presented in DISCC Workshop (*held in conjunction with MICRO'22*)
 - Pramesh Pandey, Asmita Pal, Koushik Chakraborty, Sanghamitra Roy: **Reliability and Uniformity Enhancement in 8T-SRAM based PUFs operating at NTC**, published in IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED) 2018.
 - Asmita Pal, Aatreyi Bal, Koushik Chakraborty, Sanghamitra Roy: **Split Latency Allocator: Process Variation-Aware Register Access Latency Boost in a Near-Threshold Graphics Processing Unit**, published in J. Low Power Electronics 13(3): 419-427 (2017).
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Work Experience

AMD

SANTA CLARA, UNITED STATES

Memory Systems Performance Architect

September 2025 – present

- Working on next-generation data center platforms for cloud computing and machine learning acceleration. My role focuses on workload characterization, memory subsystem performance, and microarchitecture exploration across tiered memory, scalable interconnects, and high-performance AI accelerators.

AMD

BOSTON, UNITED STATES

Research Intern

June 2022 – Sept 2022

- Characterized last-level cache (LLC) traffic for AI and HPC workloads on multicore CPU SoCs to uncover access patterns as performance bottlenecks.
- Explored reuse-distance guided heuristics and ML-based approaches to inform adaptive cache replacement strategies.
- Contributed to a US patent on dynamic cache management using per-region reuse distance estimation.

AMD

BOSTON, UNITED STATES

Research Intern

September 2020 – December 2020

- Profiled memory and LLC behavior using hardware performance counters to analyze BERT inference kernels.
- Evaluated software-hardware interaction metrics to identify performance bottlenecks and acceleration opportunities in memory-bound components.
- Generated kernel-specific traces to guide system-level optimizations targeting memory intensity and execution hotspots.

Infosys Technologies Limited

PUNE, INDIA

System Engineer

July 2014 – August 2016

- Developed an automation framework for hardware verification, streamlining debugging processes and reducing manual verification time, to enhance team productivity.
- Managed client deadlines and service delivery, coordinating with cross-functional teams.

Research Experience

Awards

- Winner of N+1 Institute X Google Reverse Pitch Competition - (AI Sustainability).
- Sarah and David Epstein Teaching Fellowship
- Student Research Grant [Awarded by Office of Diversity, Inclusion and Funding at **UW-Madison**]
- ISCA 2022 Travel Grant
- Anita Borg Grace Hopper Celebration 2019 Scholarship
- Wisconsin Distinguished Fellow - Schneider
- Rajendra Prasad and Yasoda Mani Grandhi Fellow

Research Interests & Technical Skills

Research Interests : Computer Architecture, System Performance Modeling, Memory Subsystems, Hardware-Software Co-design, Secure and Private Computing, Approximate Computing, Workload Characterization, Trace-based Simulation, NoC Security

Programming Languages : C++, Python, Bash, Verilog

Architectural Tools : ChampSim, Gem5, Garnet, PinTool, Perf, likwid

Other Libraries : Tensorflow, HELayers, OpenFHE

Relevant Coursework

Advanced Computer Architecture I and II, Introduction to Operating Systems, Neural Networks, Topics in Security and Privacy, Introduction to Programming Languages and Compilers, Embedded Computing Systems

Professional Service

Invited Talks

- Town Hall - University of Wisconsin-Madison, Dept. of Electrical and Computer Engineering, February 2025
- Computer Architecture Coordination Seminar Series - University of Central Florida, November 2023

Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture 2023 (MICRO 2023, 2022)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2022)
- Conference on Machine Learning and Systems (MLSys 2020)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)

Teaching

ECE 757 (Advanced Computer Architecture II)

Spring 2024, Spring 2025

ECE 352 (Digital System Fundamentals)

Fall 2024

ECE 5750 (High Performance Computer Architecture)

Fall 2017